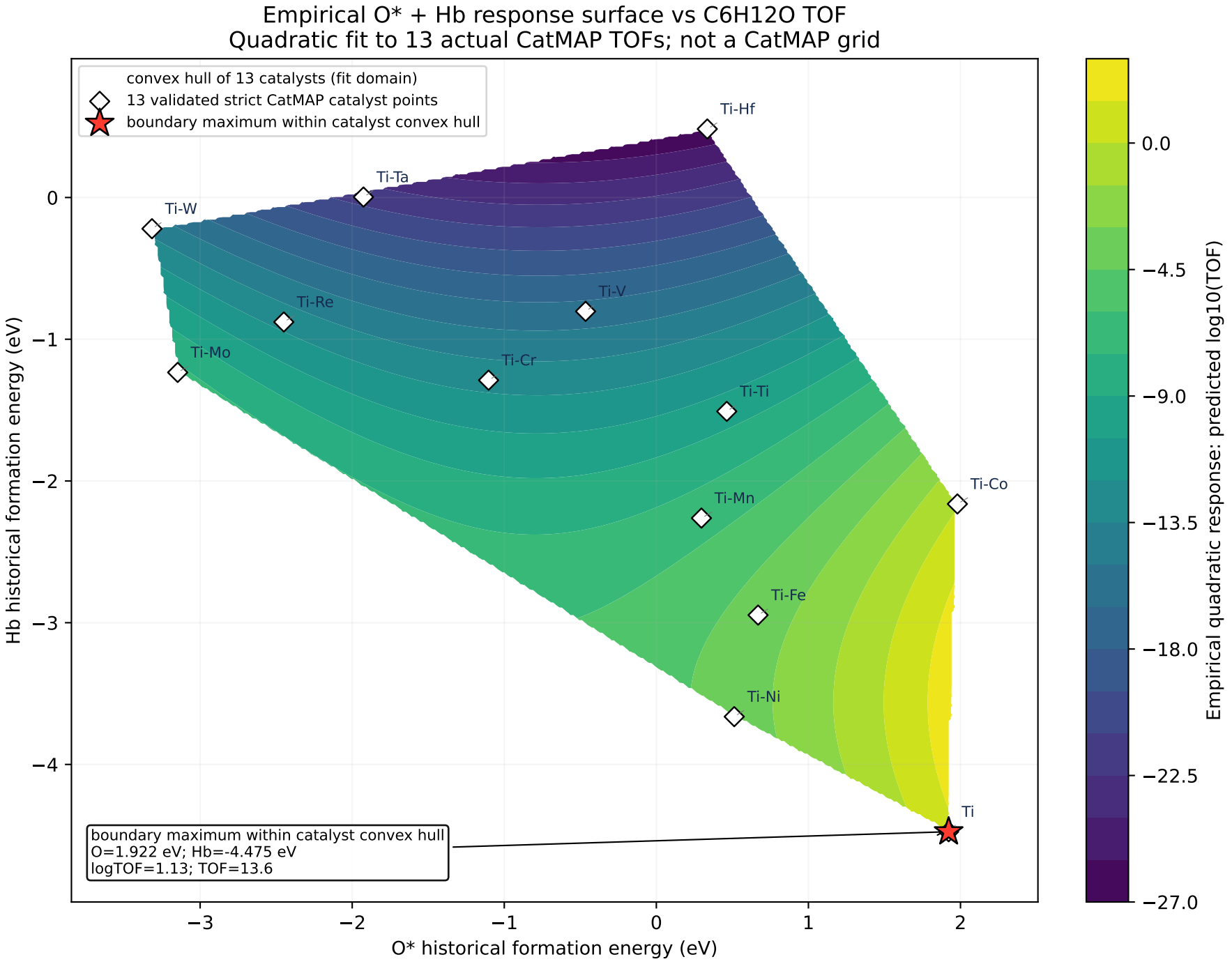

Empirical O* + Hb response surface vs C6H12O TOF
Quadratic fit to 13 actual CatMAP TOFs; not a CatMAP grid
convex hull of 13 catalysts (fit domain)
13 validated strict CatMAP catalyst points
boundary maximum within catalyst convex hull
Ti-Hf
Ti-Ta
Ti-W
Ti-V
Ti-Re
Ti-Mo
Ti-Cr
Ti-Ti
Ti-Co
Ti-Mn
Ti-Fe
Ti-Ni
Ti
boundary maximum within catalyst convex hull
O=1.922 eV; Hb=-4.475 eV
logTOF=1.13; TOF=13.6
Hb historical formation energy (eV)
O* historical formation energy (eV)
Empirical quadratic response: predicted log10(TOF)